Practical No: - 3

ON

Cryptography and Network Security - SSCS 4020

TITLE: Implementation of Hill Cipher Encryption and Decryption using Matrix Multiplication and Modular Arithmetic

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY (HONOURS.)

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Objective

To implement the Hill Cipher encryption and decryption algorithm using matrix multiplication and modular arithmetic in Python.

Theory

The Hill Cipher is a classical polygraphic substitution cipher developed by **Lester S. Hill** in 1929. It is based on linear algebra and uses matrix multiplication to encrypt blocks of letters instead of encrypting one letter at a time.

INPUT : (hill_cipher.py)

```
key = [[3, 3],
    [2, 5]]
# Inverse Key Matrix (Modulo 26)
inverse_key = [[15, 17],
        [20, 9]]
# Convert text to numbers
def text_to_numbers(text):
  return [ord(ch) - ord('A') for ch in text]
# Convert numbers to text
def numbers_to_text(numbers):
  return ''.join(chr(num + ord('A')) for num in numbers)
# Prepare plaintext
def prepare_text(text):
  text = text.upper().replace(" ", "")
  if len(text) % 2 != 0:
    text += "X"
  return text
# Encryption Function
def encrypt(text):
  text = prepare_text(text)
  nums = text_to_numbers(text)
  result = []
```

```
    for i in range(0, len(nums), 2):
        x = nums[i]
        y = nums[i + 1]
        c1 = (key[0][0] * x + key[0][1] * y) % 26
        c2 = (key[1][0] * x + key[1][1] * y) % 26
        result.extend([c1, c2])
    return numbers_to_text(result)
# Decryption Function
def decrypt(cipher):
    nums = text_to_numbers(cipher)
    result = []
    for i in range(0, len(nums), 2):
        x = nums[i]
        y = nums[i + 1]
        p1 = (inverse_key[0][0] * x + inverse_key[0][1] * y) % 26
        p2 = (inverse_key[1][0] * x + inverse_key[1][1] * y) % 26
        result.extend([p1, p2])
    return numbers_to_text(result)
# Main Program
plaintext = input("Enter Plain Text: ")
ciphertext = encrypt(plaintext)
decrypted = decrypt(ciphertext)
print("\n----------- RESULT -----------")
print("Plain Text     :", prepare_text(plaintext))
print("Encrypted Text  :", ciphertext)
print("Decrypted Text  :", decrypted)
if prepare_text(plaintext) == decrypted:
    print("Verification    : SUCCESS")
else:
    print("Verification    : FAILED")
```

OUTPUT:

```
Enter Plain Text: RISHABH

----------- RESULT -----------
Plain Text      : RISHABHX
Encrypted Text  : XWXTDFMZ
Decrypted Text  : RISHABHX
Verification    : SUCCESS
```

<u>Student Task</u>

Input:

```
key = [[3, 3],
    [2, 5]]
inverse = [[15, 17],
      [20, 9]]
text = "INFORMATION SECURITY"
# Remove spaces and convert to uppercase
text = text.upper().replace(" ", "")
# Add X if length is odd
if len(text) % 2 != 0:
  text += "X"
# Encryption
cipher = ""
for i in range(0, len(text), 2):
  a = ord(text[i]) - 65
  b = ord(text[i + 1]) - 65

  c1 = (key[0][0] * a + key[0][1] * b) % 26
  c2 = (key[1][0] * a + key[1][1] * b) % 26
  cipher += chr(c1 + 65)
  cipher += chr(c2 + 65)
# Decryption
plain = ""
```

```
for i in range(0, len(cipher), 2):
    a = ord(cipher[i]) - 65
    b = ord(cipher[i + 1]) - 65
    p1 = (inverse[0][0] * a + inverse[0][1] * b) % 26
    p2 = (inverse[1][0] * a + inverse[1][1] * b) % 26
    plain += chr(p1 + 65)
    plain += chr(p2 + 65)
print("Original Plain Text :", "INFORMATION SECURITY")
print("Plain Text          :", text)
print("Encrypted Text      :", cipher)
print("Decrypted Text      :", plain)
if text == plain:
    print("Verification        : SUCCESS")
else:
    print("Verification        : FAILED")
```

OUTPUT:

```
Original Plain Text : INFORMATION SECURITY
Plain Text          : INFORMATIONSECURITYX
Encrypted Text      : LDFCJQFROIPMSSHVDHLH
Decrypted Text      : INFORMATIONSECURITYX
Verification        : SUCCESS
```

Advantages and Disadvantages of Hill Cipher

Advantages	**Disadvantages**
Encrypts multiple letters (blocks) at a time, making it stronger than simple substitution ciphers.	The key matrix must be invertible modulo 26; otherwise, decryption is not possible.
Uses matrix multiplication, making encryption more complex than monoalphabetic ciphers.	Not secure against modern cryptanalysis and known-plaintext attacks.
Provides better security than basic substitution ciphers.	Key generation and matrix calculations are more complex than simple ciphers.
Fast and easy to implement for small messages.	Requires padding (such as adding **X**) when the plaintext length is odd.
Helps in understanding matrix operations and modular arithmetic in cryptography.	Mainly suitable for educational purposes and not recommended for real-world secure communicat

Viva Questions

Q1. What is Hill Cipher?
Answer: Hill Cipher is a polygraphic substitution cipher that uses matrix multiplication and modular arithmetic for encryption and decryption.

Q2. Which mathematical operation is used?
Answer: Matrix multiplication modulo 26.

Q3. Why is modulo 26 used?
Answer: Because the English alphabet contains 26 letters.

Q4. What is the key matrix?
Answer: The matrix used for encryption. In this practical:

[3 3]

[2 5]

Q5. Why is the inverse matrix required?
Answer: It is required to decrypt the ciphertext back into the original plaintext.

Q6. What happens if the plaintext length is odd?
Answer: An extra letter (usually **X**) is added to make the length even.